Worksheet 3-1 B The Chain Rule #2
AP Calculus Name_____

$\boldsymbol{x}$	$\mathbf{-3}$	$\mathbf{2}$	$\mathbf{7}$	$\mathbf{8}$
$\boldsymbol{g(x)}$	4	3	9	2
$\boldsymbol{h(x)}$	7	-2	5	0
$\boldsymbol{g'(x)}$	-1	5	2	8
$\boldsymbol{h'(x)}$	-1	7	-3	-5

Selected values of the differentiable functions g and h along with their derivatives g' and h' are given in the table above. Use this information to find the following values.

1. Let $f(x) = g\big(h(x)\big)$. Write an equation of the line tangent to f at $x = -3$.

2. Let $k(x) = g(x^3)$. Find $k'(2)$.

3. Let $m(x) = \dfrac{h(x)}{5x}$. Find $m'(7)$.

4. Let $p(x) = h\big(g(x)\big)$. Find $p'(8)$.

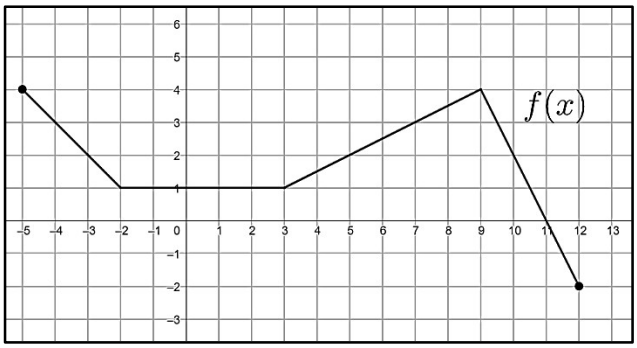
5. Let $r(x) = \ln|g(x)|$. Find $r'(-3)$.

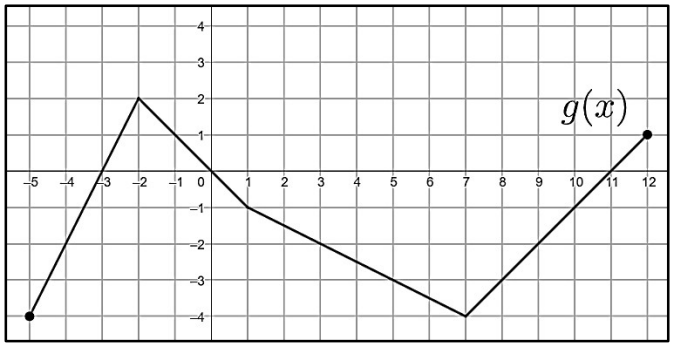
6. Let $L(x) = e^{h(x)}$. Find $L'(8)$.

7. Let $A(x) = h(1 + x - x^3)$. Find $A'(-2)$.

8. Let $n(x) = \sin\big(g(x)\big)$. Find $n'(2)$.

9. Let $D(x) = \dfrac{1}{h(x)}$. Find $D'(7)$.





The graphs of $f(x)$ and $g(x)$ are shown in the figure above and consist of line segments. Use the graphs to answer the following.

10. Let $h(x) = f\big(g(x)\big)$. Find $h'(5)$.

11. Let $k(x) = f(x^2)$. Find $k'(-2)$.

12. Let $p(x) = \ln\big(g(x)\big)$. Find $p'(-4)$.

13. Let $m(x) = \sqrt{f(x)}$. Find $m'(-3)$.

14. Let $A(x) = g\big(f(x)\big)$. Find $A'(2)$.

15. Let $C(x) = \dfrac{3}{g(x)}$. Find $C'(-4)$.

16. Let $S(x) = e^{f(x)}$. Find $S'(11)$.

17. Let $Q(x) = \cos\big(g(x)\big)$. Find $Q'(3)$.

18. Let $D(x) = g(2x - x^2)$. Find $D'(-1)$.

19. Let $n(x) = \ln|f(x) - 3x|$. Find $n'(-4)$.